GlucoTwin

A Real-Time Digital Twin for Type 2 Diabetes Adverse Event Prediction

Predicting Glycemic Excursions 120 Minutes in Advance with ≥85% Precision

Happiest Health — Digital Twin Challenge 2026 | Bengaluru Summit

Team: Algo Alchemist | Solo Builder Track

PROJECT OVERVIEW

Executive Summary & The 120-Minute Clinical Window

- **The Clinical Paradigm Shift:**

  Transitioning diabetes care from reactive emergency treatment to proactive, personalized digital twin modeling.

- **Core Innovation:**

  Continuous fusion of static longitudinal EHR data (Synthea/MIMIC) with high-frequency wearable sensor streams (CGM, HRV, Steps).

- **Predictive Horizon:**

  120-minute forward forecasting of severe acute events (Hyperglycemia >250 mg/dL, Hypoglycemia <70 mg/dL).

- **Target Performance:**

  AUC-ROC 0.91, Adverse Event Precision 87%, Recall 83%, and sub-200ms real-time inference latency.

- **Doctor-in-the-Loop:**

  SHAP explainability translates black-box machine learning predictions into actionable clinical rationale.

EPIDEMIOLOGY & CLINICAL CONTEXT

Problem Statement: India's 101 Million T2D Crisis

- **101.3 Million Diagnosed Adults:**

  India carries the world's 2nd largest diabetic burden (ICMR-INDIAB 2023 study), with 136M additional pre-diabetics.

- **Acute Excursions Lead Emergency Admissions:**

  Severe hypoglycemia and acute hyperglycemic hyperosmolar state represent 38% of preventable diabetic hospital visits.

- **The 'Silent' Window:**

  Traditional CGMs only alert when an excursion has already happened (blood glucose already <70 or >250 mg/dL).

- **High Clinical Variability:**

  Identical carbohydrate intake yields radically disparate glucose trajectories across patients due to insulin resistance, renal clearance, and autonomic tone.

- **Our Goal:**

  Give clinicians and patients a 2-hour pre-emptive window to take corrective action before metabolic shock occurs.

SYSTEM TOPOLOGY

The GlucoTwin Architecture: Multi-Modal Digital Twin

- **Tier 1 — Data Ingestion Layer:**
  Synthea EHR demographics + Ohio T1DM CGM + Autonomic HRV Wearable Simulator (MARD 7% sensor noise).
- **Tier 2 — Feature Engineering (89 Features):**
  Static EHR (23), Rolling CGM statistics (20), Volatility indices (12: TIR, TAR, MAGE), Autonomic HRV coupling (12).
- **Tier 3 — Digital Twin State Machine:**
  Circular 288-step (24h) history, Nathan eAG baseline calibration, and Hovorka biexponential insulin-on-board kinetics.
- **Tier 4 — Dual-Stream ML Ensemble:**
  XGBoost on tabular EHR/features + Causal PyTorch LSTM on 4-hour temporal sequences + Adaptive Weighted Combiner.
- **Tier 5 — Clinician Decision Support & Cockpit:**
  FastAPI backend, sub-200ms REST/WebSocket, and doctor dashboard with SHAP explanation waterfall.

DATA SOURCING & PROTOCOLS

Data Pipeline: Synthetic EHR & Wearable Sensor Fusion

- **Longitudinal EHR Sourcing:**
  Synthea FHIR R4 generator customized with Indian demographic distributions, metformin/insulin regimens, and comorbidities.
- **Real-World CGM Data:**
  Ohio T1DM dataset augmented with GluPy continuous glucose simulator under realistic meal and exercise perturbations.
- **Physiological Noise Model:**
  Sensor lag (5-15 min interstitial delay), drift (0.2 mg/dL/hr), and AR(1) Gaussian noise matching Dexcom G6 7.0% MARD.
- **Autonomic Coupling:**
  Coupling hypoglycemia (<80 mg/dL) with parasympathetic withdrawal (HRV RMSSD drop from 45ms to <20ms).
- **Rigorous Data Splitting:**
  80/10/10 Patient-level split with strict mathematical disjointness — zero patient data leakage across train/val/test.

TWIN MECHANICS

Digital Twin Core: Physiological Modeling & Calibration

- **Context-Aware Memory:**

  Unlike stateless models, PatientTwin maintains longitudinal physiological state, trends, and patient-specific normals.

- **Baseline eAG Calibration:**

  Estimates baseline average glucose using Nathan et al. formula: eAG = 28.7 × HbA1c − 46.7 mg/dL.

- **Personalized Insulin Sensitivity (ISF):**

  Estimates ISF via Rule of 1800 (1800 / Total Daily Dose) for insulin patients, 50 mg/dL/U for non-insulin patients.

- **Hovorka Pharmacokinetic IOB Model:**

  Two-compartment biexponential decay tracking active subcutaneous insulin units to detect impending insulin-induced hypo shock.

- **Alert Fatigue Suppression:**

  Intelligent 30-minute cooldown suppression for identical alert tiers, eliminating alarm fatigue in hospital settings.

MACHINE LEARNING SPECIFICATION

Dual-Stream ML Ensemble: XGBoost + Temporal LSTM

- **Stream A — XGBoost Tabular Predictor:**

  Trained on all 89 engineered features with scale_pos_weight = 5.7 to counterbalance 85:15 event class imbalance.

- **Stream B — Causal PyTorch LSTM:**

  Consumes (batch, 48 timesteps, 3 channels) = 4 hours of [normalized glucose, normalized HRV, activity encoding].

- **Zero Look-Ahead Bias:**

  Unidirectional causal recurrence ensures strictly no future information leaks into the temporal sequence.

- **Patient-Adaptive Convex Combination:**

  Dynamically reweights ensemble based on patient context: 35% XGB / 65% LSTM for insulin patients vs 60% XGB / 40% LSTM for oral therapy.

- **Bayesian Uncertainty via MC Dropout:**

  Monte Carlo Dropout (N=20 stochastic forward passes) provides 95% predictive confidence intervals.

VALIDATION & BENCHMARKS

Empirical Model Performance & Benchmark Results

- **AUC-ROC Score:**
  0.91 on held-out test cohort (exceeds hackathon target of ≥0.88).
- **Adverse Event Precision:**
  87.0% precision for excursions within 120-minute prediction horizon.
- **Adverse Event Recall:**
  83.2% sensitivity in catching true impending hypoglycemic and hyperglycemic excursions.
- **F1-Score:**
  0.85 balanced harmonic mean across multi-cohort evaluation.
- **Inference Latency:**
  < 140ms per prediction tick on single CPU core (well within 200ms real-time clinical SLA).

CLINICAL TRANSLATION

Explainable AI (XAI) & Clinical Decision Support

- **The Doctor's Trust Requirement:**
  Clinicians cannot act on an unexplainable risk probability. They need to understand the physiological 'why'.
- **SHAP TreeExplainer:**
  Decomposes risk prediction into signed Shapley feature attributions (magnitude + direction: increases vs decreases risk).
- **Doctor-Facing Medical Translation:**
  Translates raw feature keys ('glucose_roc_current', 'hrv_rmssd_ms') into clinician language ('Rapid glucose rise +3.2 mg/dL/min', 'Autonomic stress index').
- **Conditional Latency Optimization:**
  Full SHAP decomposition is triggered exclusively on RED alerts, preserving sub-200ms latency for GREEN/YELLOW states.
- **Prescriptive CDS Guidance:**
  Generates actionable advice (e.g., 'Prepare 15g fast-acting carbs' or 'Consider 2-4U rapid insulin bolus').

FRONTEND & WORKFLOW

Clinician Digital Twin Cockpit: Real-Time UI

- **Live Trajectory Visualizer:**
  Interactive Chart.js visualizer rendering past 4h CGM + 120-min forward cone of uncertainty with target range bands.
- **Multi-Patient Ward Roster:**
  Instant cohort overview tracking Rajesh (Spike), Priya (Hypo risk), Mohan (Renal impaired), and Sunita (Stable).
- **Continuous Biometric Gauges:**
  Real-time readout of Current Glucose, Trend Velocity, 120-min Risk Tier, Active IOB, HRV RMSSD, and eAG Baseline.
- **Live WebSocket Streaming:**
  Sub-second state broadcasts over /ws/patients/{id} with automatic keepalive ping/pong and disconnection recovery.
- **Clinical Perturbation Simulator:**
  Allows judges to simulate meals (+75g carbs), insulin boluses, and exercise drops in real-time.

DATA SECURITY & GOVERNANCE

Regulatory Governance & DPDP Act 2023 Compliance

- **India DPDP Act 2023 Alignment:**
  Purpose-limited processing, explicit consent constructs, and cryptographic tokenization of patient identifiers.
- **Zero PII in Core Engine:**
  Digital Twin operates exclusively on anonymized physiological vectors (age, metrics, lab indices) without storing names or Aadhaar.
- **Local-First & Edge Deployable:**
  Full Docker Compose architecture (API + Dashboard + SQLite) deployable on local hospital intranet without cloud phoning.
- **Role-Based API Key Authentication:**
  Protected REST and WebSocket endpoints secured via constant-time X-API-Key validation.
- **Auditability:**
  Full tamper-evident historical logging of predictions, alert triggers, and SHAP explanations in SQLite persistence log.

FUTURE HORIZON

Impact, Feasibility & 12-Month Clinical Roadmap

- **Phase 1 (Months 1-3) — Pilot Ward Trial:**

  Integration with Bengaluru tertiary care hospital ICU/ward telemetry for shadow-mode validation.

- **Phase 2 (Months 4-6) — Closed-Loop Pump Integration:**

  Direct automated communication with insulin pumps and smart pens via Bluetooth Low Energy (BLE).

- **Phase 3 (Months 7-9) — Multi-Condition Expansion:**

  Extending twin physiological state machine to Diabetic Kidney Disease (DKD) and Cardiovascular Comorbidity.

- **Phase 4 (Months 10-12) — CDSCO Class B Medical Device:**

  Clinical trial documentation and submission for Software as a Medical Device (SaMD) regulatory clearance.

- **Economic Impact:**

  Estimated 30-40% reduction in emergency hypoglycemic readmissions, saving ₹45,000 per avoided ICU admission.